Challenge: Sumo Ring Push-Out

Points: 250 • Estimated Time: 90 Minutes

Challenge Scope

The robot must stay inside a circular ring while locating and pushing a passive target object outside the boundary.

Instructions & Engineering Hints

• Welcome to the **Sumo Ring Push-Out Challenge**! Your mission is to program and configure your autonomous robot to locate a passive target object within a circular arena and push it entirely outside of the boundary line while ensuring your own robot never crosses or touches that boundary.

• **Architectural Hint - Sensors:** Think carefully about how your robot will perceive the edge of the ring versus the target object. Placing downward-facing light or color sensors near the front or bottom of your chassis can help detect contrasting boundary lines (such as a black ring on a white floor, or vice versa).

• **Architectural Hint - Navigation & Locating:** How will your robot find the target? Consider mounting an ultrasonic or distance sensor at the front to sweep the arena, or design a physical 'cowcatcher' or plow scoop that passively guides the target as the robot moves forward.

• **Strategic Approach:** Balance your drive speeds and logic loops. If your robot drives too fast, it might not stop in time when it detects the boundary line. Implement conditional statements (if-else logic) so that retreating or turning away from the boundary takes absolute priority over pushing the target.